Courseware prospectus

Using Data to Drive Business Performance

Research and proposal review · October 2026

What is this book about?

Courseware for undergraduate business students that connects the way data is organized, queried and managed to the decisions an organization makes.

In this course, students learn how to understand data, organize it, store it in a relational database, ask questions of it in SQL, check the answers, analyze and visualize the results in Power BI, and make a decision. AI appears throughout as an assistant whose work students learn to check.

Textbook

A textbook, interactive reader, guided practice, independent labs, reading assessments, and a semester project are designed to work as one course. The proposed next edition has 16 chapters in five parts.

Instructor materials

The courseware also includes materials for instructors:

Review questions · Terms Treasury · Answer keys · Automatic graders · Rubrics · Slides · Lesson plans · Instructor dashboard

Who it is for

Business core courses	The required data or information systems course for accounting, finance, marketing, management and operations students, who need to query, check and present data.
MIS and analytics majors	A full database course: modeling, normalization, SQL through window functions, administration, warehousing and business intelligence.

Adjacent courses	Accounting information systems, business analytics foundations, and bridge courses in IS master's programs.

Why and how it was built

I wrote this book because, after more than six years of teaching this course to over 4,000 students with materials from all the leading publishers, I still could not find a textbook that matched my needs.

Available textbooks were either too technical, too thin on hands-on practice, or missing crucial content. As cloud databases, analytics and visualization tools, and AI entered the workplace, the gap became stark. I often assigned at least two books and filled the rest of the gaps myself.

The market and curriculum research below helped define a focused opportunity: courseware that brings database foundations, hands-on work and business interpretation together.

Based on my experience and on the research, I concluded that the content, the tools and the speed of revision all had to change. So I built the courseware myself, chapter by chapter, over a year of work with colleagues, and tested it live in the classroom.

Ready now

- The full textbook, live as an interactive digital edition and in use at UAlbany
- A Let's Build, an interactive lab and an auto-graded quiz in every chapter
- Student accounts, automatic grading and an instructor dashboard

Open the live courseware

Planned for the next edition

- The research-based outline in the next section
- Runnable PostgreSQL inside the reader, with practice workspaces
- A tested MySQL route alongside the Access route
- Automated feedback on SQL and lab answers (prototype in testing)

Classroom history and usage figures are supplied by the author; they describe development and use, rather than independently measured learning gains.

Market research

How the market research was done

What we reviewed	Federal degree statistics (NCES and IPEDS, through Data USA), peer-reviewed studies of business and IS curricula, AACSB accreditation data, U.S. Bureau of Labor Statistics projections, the World Economic Forum's employer survey, and an analysis of 855 U.S. data analyst job postings. Our own syllabus research, below, adds the textbook and AI-policy findings.
How	We retained published figures with their source years, separated degree counts from course and program evidence, and used two combinations as illustrative reach proxies. Each proxy is shown with its assumptions and requires publisher validation.
What we looked for	How many students take a course like this each year, which programs require it, what employers ask for, and whether any textbook already owns the market.

Finding 1: a large, required course

About 375,400 degrees a year	**Business bachelor's degrees awarded in the U.S.** About one in five of all bachelor's degrees (2021-22).Source: NCES [1]
88% of business schools	**Require an information systems course of every business student** In a study of 61 business schools, 89% of them AACSB-accredited, 85% required one IS course in the core and 3% required two.Source: Fustos et al. [2]
12,092 degrees in 2024	**Management information systems degrees awarded** Up 3.4% from 2023.Source: Data USA [3]
87.3% of IS programs	**Offer a data and information management course** 344 of 394 U.S. undergraduate IS programs.Source: Hwang, Ma and Wang [4]
285 of 535 AACSB schools	**U.S. AACSB-accredited schools with an information systems program** A further 109 offer an undergraduate business analytics program.Source: Zheng et al. [5]
82% of IS master's programs	**Require a data management course** 61 of 74 programs at AACSB schools: a graduate audience for the same material.Source: Yang [6]
81,835 degrees in 2024	**Accounting degrees awarded, an adjacent audience** Accounting information systems courses cover much of the same ground.Source: Data USA [7]

1,098 schools	**AACSB-accredited business schools worldwide** In 71 countries, as of September 2026.Source: AACSB [8]

Two routes, one book

A commonality we found: **most business schools require an information systems course of every business student, and most MIS programs require a database course.** The proposed routes address these two settings; fit must be reviewed against each course's outcomes and available teaching time. In round numbers:

About 330,000 Illustrative reach proxy	**Business students who reach a required IS course** About 375,400 business bachelor's degrees × the 88% of business schools that require the course.Source: Our estimate from [1], [2]
About 10,600 Illustrative reach proxy	**MIS students in programs with a database course** 12,092 MIS awards across all levels × the 87.3% of IS programs that offer the course.Source: Our estimate from [3], [4]

Planning assumptions: these reach proxies multiply degree counts by historical shares of institutions or programs. They are not enrollment-weighted course-seat estimates. The MIS and accounting totals cover all award levels, not bachelor's degrees alone. The routes overlap and must not be added together. Publisher enrollment and adoption data are needed to estimate addressable annual seats.

Finding 2: a varied textbook landscape

60 titles	**Textbooks named in the syllabi we reviewed** No title appeared at more than three institutions in the earlier 60-title catalogue. Instructor-made materials also appear. This indicates variety within the reviewed collection, rather than national market share.Source: Syllabus research, below
2021 last revision	**Cengage's applied business-database title** Concepts of Database Management, 10th edition, is built on Access 2019 and SQL Server 2019. It has served the course well, and the tools and topics have moved on since.Source: Cengage [13]

Student fees and adoption opportunity

Existing product / format	Listed U.S. price	Access or format
Database Systems 15e eBook	US$71.99	Access through January 22, 2027
Database Systems 15e eBook	US$98.99	Access through July 21, 2027
Database Systems 15e paperback	US$272.95	Print purchase
MindTap for Database Systems 15e	From US$149.00	Digital courseware; product / term selection applies

Segment	Likely need	Validation focus	Adoption opportunity
Primary Applied business database / MIS data management	Relational thinking, SQL, design, implementation, and business interpretation in one coherent learning path.	Course outcomes, annual sections and enrollments, required engine, incumbent materials, and switching criteria.	Full courseware adoption with readings, guided practice, independent labs, assessment, and instructor resources.
Business analytics foundations	Data preparation, trustworthy SQL measures, and reporting.	Required design depth, analytical prerequisites, and reporting platform.	A selected chapter / lab route that supplies the database foundation for analytics.
Business core / accounting information systems	Data meaning, information systems, controls, reporting, and a bounded project.	Which modules fit the existing course and how much SQL / design is expected.	Modular adoption or a companion pathway, subject to course fit.

Illustrative adoption scenario	Annual sections per institution	Students per section	Annual student seats	Validation focus
10 institutions	2	30	600	Recruit a small set of suitable courses; establish required tools and sample-unit fit.

Illustrative adoption scenario	Annual sections per institution	Students per section	Annual student seats	Validation focus
25 institutions	2	30	1,500	Confirm repeat adoption, instructor support needs, and section-level enrollments.
50 institutions	2	30	3,000	Validate capacity, delivery costs, distribution, and broader portfolio fit.

Student prices were retrieved October 25, 2026 and vary by format, access period and institutional arrangement. Adoption scenarios are arithmetic examples, not forecasts. A price for this proposed edition has not been set.

Main conclusion The evidence supports exploring two defined course routes: the required business-core data / IS course and the applied MIS database course. Validate their outcomes, tools, annual sections, student seats and switching criteria separately.

Curriculum evidence

Why and how the syllabus research was done

Why	To check the outline against what courses across the country actually teach: which topics, which tools, which textbooks, how they assess, and what they say about AI.
What we collected	63 syllabi and course documents: syllabi, catalog descriptions and course schedules for database and business-data courses, mostly undergraduate, from the public websites of 31 universities. Retrieved October 1, 2026.
Why 63 documents but 31 institutions	Many institutions contribute more than one course, and some courses have more than one document. The 63 documents describe 54 distinct courses at 31 institutions. Of the 54 courses, 19 are business-database or business-core data courses, the closest match to this book.
How we read them	Archived documents and the earlier audits were reviewed for topics, platforms, textbooks, assessment and AI policies. Each set of findings retains its own sample and denominator.

Institutions analyzed

- Baruch College
- Boston University
- Brooklyn College
- CUNY City Tech
- Florida International University
- George Mason University
- Georgia Tech
- Hofstra University
- Lafayette College
- Michigan Dearborn
- NYU Stern
- Northeastern University
- Penn State
- Rutgers University
- SUNY Canton
- SUNY Farmingdale
- SUNY New Paltz
- SUNY Oswego
- SUNY Tompkins Cortland
- San Jose State University
- Stony Brook University
- Syracuse University
- Temple University
- UMass Amherst
- UMass Dartmouth
- UNLV

- UT Austin
- UT Dallas
- University at Albany
- University at Buffalo
- University of Georgia

What the courses teach, based on our analysis

Explicit topic mentions in the earlier 13-course catalogue benchmark. These are lower bounds from course descriptions, not topic frequencies across the 63-document archive.

Design and data modeling	100%
SQL and querying	77%
Hands-on building	77%
Admin, security, transactions	46%
Normalization	31%
ER modeling	31%
BI and analytics	31%
NoSQL	15%
AI	8%
Cloud computing, APIs, ETL	0%

Which databases they use

Role-coded evidence distinguishes practical use, named options and software-list mentions. Counts overlap across the 54 selected course families.

Platform	Practical use	Option	Listed only	Course families	Institutions	Direct / business-core
MySQL	10	3	2	15	11	6
Microsoft Access	10	0	2	12	8	9
Oracle	4	2	2	8	7	2
Microsoft SQL Server	3	3	0	6	4	2
PostgreSQL	1	4	0	5	4	0
SQLite	0	1	1	2	2	0
Google BigQuery	3	0	0	3	2	1

How they assess, and what they say about AI

Assessment	The proposed chapter labs and feedback make guided practice explicit before cumulative assessment.
AI	Twelve of 18 primary offerings dated 2024-2026 include inline student AI-assistance provisions. Five of those 18 offerings document no student AI rule; one refers to linked policy whose contents were unavailable.

How did the findings shape the book?

Design and data modeling Named in all 13 catalogue benchmark descriptions.	**A two-chapter design block:** requirements to ER diagrams, then normalization.Ch 7Ch 8
SQL and hands-on building Each appears explicitly in 10 of the 13 benchmark course descriptions (77%).	**Three SQL chapters,** a build-and-test chapter, and a Let's Build in every chapter.Ch 5Ch 6Ch 9Ch 10

Multiple platforms Access has the largest documented footprint in the Direct / Business-core subset; MySQL leads across all reviewed course families. The archive also includes Oracle, SQL Server, PostgreSQL and other systems.	**Teach SQL in the PostgreSQL dialect,** with a portable core and explicit dialect differences, with tested MySQL and MS Access routes on the same cases, and Supabase as the proposed hosted PostgreSQL environment.App AApp BApp C
BI and analytics Named in 4 of 13 benchmark descriptions (31%).	**Analytics, reporting and Power BI chapters** close the loop from query to decision.Ch 12Ch 13Ch 14
AI and cloud AI / ML appears in 1 of 13 benchmark descriptions; cloud / APIs / ETL in none. This does not measure their prevalence across the full archive.	**AI verification throughout the course,** and a dedicated chapter on AI, cloud computing and data warehouses.Every chapterCh 15
Staged projects Most courses run staged, multi-phase projects.	**One staged project through the semester,** with graded checkpoints and an individual demonstration at the end.Studio ACh 11Ch 16

Method: 63 archived documents, 31 institutions, 54 grouped course families; topic percentages use a separate 13-course catalogue benchmark, and the AI-policy review uses 18 recent primary offerings. The searchable register below retains source dates and course roles.

From evidence to design

Institution	Course	Evidence date / level	Documented emphasis	Implication for this proposal
Rutgers	33:136:470	Fall 2026	SQL early, design/normalization, operational topics and a staged project.	Supports use-before-design and a cumulative project.
Baruch	CIS3400	Fall 2026 draft	Business requirements, ER modeling, normalization, SQL and an Access application.	Preserve application thinking and a substantive Access companion.
NYU Stern	TECH-UB26	Spring 2026	BigQuery SQL, cleaning, joins, aggregation, subqueries and window queries; ER concepts.	Supports browser/cloud SQL and an analytics-facing route.
Temple	MIS2502	Spring 2026	SQL, schemas, normalization, Python/JSON, data movement and analytics.	Connect query work to preparation and interpretation.
Georgia Tech	MGT4058	Fall 2026	Design, relational mapping, normalization, SQL and BI/warehouse concepts.	Protect the database core while connecting to reporting.
Boston University	MET-CS669	Spring 2026 · graduate	Design, SQL, operations and a project developed across six iterations.	Staged project structure is useful; graduate technical depth remains optional.
UT Dallas	ITSS4300	Spring 2026	SQL, modeling, normal forms, CTEs, windows, BI and security.	Provide advanced SQL extensions without removing essential design.

Institution	Course	Evidence date / level	Documented emphasis	Implication for this proposal
University of Georgia	MIST4610	Summer 2020	Modeling, MySQL, query work and visual answers to business questions.	Supports a query-to-communication chain; platform evidence is historical.

Industry research

What we researched

A 2026 analysis of 855 relevant U.S. data analyst job postings, U.S. Bureau of Labor Statistics projections, and the World Economic Forum employer survey.

What employers ask for

We looked at what employers ask for in the skills this book teaches.

52.9% of job postings	**U.S. data analyst job postings that ask for SQL** Of 855 postings analyzed. Python appears in 31.2%, Power BI in 29.0% and Tableau in 26.2%.Source: 365 Data Science [9]
+35% growth, 2025 to 2035	**Projected growth in data scientist jobs** Against 3% for all occupations. Median pay: $120,230 a year.Source: U.S. Bureau of Labor Statistics [10]
+9% growth, 2025 to 2035	**Projected growth in database architect jobs** Median pay: $139,500 a year.Source: U.S. Bureau of Labor Statistics [11]
About two-thirds of employers worldwide	**Plan to hire for AI skills** AI and big data are the fastest-growing skills employers report.Source: World Economic Forum [12]
SQL	52.9%
Python	31.2%
Power BI	29.0%
Tableau	26.2%

Share of 855 U.S. data analyst job postings that list each skill.

Evidence	Finding	Courseware implication
BLS: data scientists, 2025-35	**35% projected growth** , compared with 3% for all occupations; about **24,800 openings annually** . [L1]	Connect data preparation and queries to interpretation, communication, and decisions.
BLS: database architects / administrators, 2025-35	Architects: **9%** growth. Administrators: **0%** . Combined: about **7,300 openings annually** . [L2]	Retain modeling, quality, permissions, backup, and system judgment alongside analytics.
WEF: Future of Jobs Report 2025	Survey of **1,000+ employers** , representing **14 million+ workers** across **55 economies** ; roughly **7 in 10** identify analytical thinking as essential. [L3]	Require students to explain a result, test its reliability, and justify an action.
WEF: skills outlook to 2030	AI and big data lead the skills employers expect to grow fastest. [L3]	Integrate responsible AI use, verification, data provenance, and limits across the learning experience.

Main conclusion SQL, reporting and judgment belong together. Students need to produce a result, establish that it is reliable, and communicate what an organization should do with it.

Job-posting shares describe this sample, not all U.S. hiring. BLS openings include replacement hiring. Global employer expectations are distinct from U.S. job-posting requirements.

Textbook research

How the textbook research was done

We reviewed the leading textbooks used in the courses we studied and compared their tables of contents, structure, tools and audience, chapter by chapter.

The textbooks reviewed

Database Systems: Design, Implementation, and Management

Coronel and Morris · Cengage, 15th edition (©2027)

Chapters	16 in 6 parts

SQL starts	Chapter 4, before ER modeling
Tools	MindTap MySQL labs; Oracle, SQL Server, Access
Best fit	IS and MIS majors who need depth

Concepts of Database Management

Friedrichsen et al. · Cengage, 10th edition (2021)

Chapters	9
SQL starts	Chapters 2 to 3
Tools	Access 2019, SQL Server 2019; MindTap and SAM
Best fit	Applied business courses

Database Concepts

Kroenke, Auer, Vandenberg and Yoder · Pearson, 10th edition

Chapters	7, with more online
SQL starts	Chapter 3
Tools	Access, MySQL 8.0, Azure; running projects
Best fit	Short, Access-first courses

Modern Database Management

Hoffer, Topi, Venkataraman and Bala · Pearson, 14th edition

Chapters	13
SQL starts	Chapter 5
Tools	Oracle, SQL Server, MySQL; covers analytics, cloud and AI

Best fit	Managerial MIS courses

Database Systems with SQL and zyLabs

Winsberg et al. · zyBooks

Chapters	12 interactive chapters
SQL starts	Early
Tools	MySQL in the browser
Best fit	Practice-heavy courses

Murach's MySQL

Murach · 4th edition

Chapters	Query-first workbook
SQL starts	Chapter 1
Tools	MySQL Workbench, AWS
Best fit	SQL skills courses

Using Data to Drive Business Performance

Dvir · proposed next edition

Chapters	16 in 5 parts
SQL starts	Chapter 5, after data literacy
Tools	PostgreSQL and Supabase; MySQL and Access routes; Power BI
Best fit	Business-core and MIS courses

Publisher pages: Coronel and Morris 15e; Concepts of Database Management 10e; Database Concepts 10e; Modern Database Management 14e.

Chapter by chapter, in order

We coded each chapter of the four main textbooks by its main topic to see which topics the books have in common and in what order they teach them. Each square is one chapter. The last row is the proposed outline of this book.

Coronel and Morris Cengage, 15th ed.	1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10 · 11 · 12 · 13 · 14 · 15 · 16
Concepts of Database Mgmt. Cengage, 10th ed.	1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9
Database Concepts Pearson, 10th ed.	1 · 2 · 3 · 4 · 5 · 6 · 7
Modern Database Mgmt. Pearson, 14th ed.	1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10 · 11 · 12 · 13
This book Proposed	1 · 2 · 3 · 4 · 5 · 6 · M · 7 · 8 · 9 · 10 · 11 · 12 · 13 · 14 · 15 · 16

M marks the midterm Integration Studio. Topic classification is ours, from publisher tables of contents.

What it tells us: the four compared sequences share relational foundations, SQL and design. The proposed edition keeps that core and extends its opening business / data-literacy sequence into dedicated analysis, reporting, Power BI and capstone work.

What the leading books do well, and this book keeps

• **The relational core:** keys, relationships, ER modeling and normalization to third normal form

• **Early, hands-on SQL,** as in Coronel 15th edition, Kroenke and zyBooks

• **Administration, transactions and security** as a full chapter

• **Warehousing, star schemas and NoSQL** as the bridge to analytics

Where they stop, and this book continues

• **The four compared chapter sequences open with database foundations rather than a three-chapter business-performance, data-literacy and information-organization sequence.** This book does, in Chapters 1 to 3.

• **A dedicated Power BI studio is a proposed distinction in this comparison,** alongside analytics, visual reporting and a capstone decision memo. Current competing texts already cover aspects of BI and analytics.

• **The proposed learning design makes AI verification a recurring practice.** This book has AI verification throughout the course.

• **Courses and books support different database products.** This proposal keeps PostgreSQL central and calls for tested MySQL and Access routes using the same cases.

zyBooks reference ↗

Murach reference ↗

Proposed courseware

Things that shaped the structure of this courseware

Seven years in the classroom	I have taught the introductory MIS course at the University at Albany for seven years, to more than 4,000 students. The chapter labs and quizzes were developed and refined across those semesters.
More than ten years of editing	Before I became a professor, I spent over ten years as a magazine editor. That shaped the writing: clear, visual and organized around the reader, at a level business undergraduates can follow.
Extensive market research	I reviewed and analyzed the main textbooks used in these courses, surveyed what universities across the country teach, and compared 63 syllabi collected from 31 institutions. The outline draws directly on that work. See the research below.
Full control and quick iteration	Because I built the platform, tools, examples and AI guidance can change each semester instead of each edition. After the summer test and the current deployment with 500 students, I collect live feedback from students and revise continuously.

The course arc

The proposed 16-chapter edition separates administration, analytics, visualization and Power BI, and gives the capstone its own chapter. Semester workload and the two course routes remain open to publisher review.

Data → Tables → Relationships → Queries → Analytics → Decisions

The following outline is based on the detailed research presented below.

The book at a glance

A preface, **16 chapters in five parts** , a midterm review week, a closing chapter on careers and seven appendices. It offers a proposed route for a 14- to 16-week semester, subject to course-level workload review. Each part below opens with its main goal, and each chapter lists its sections, its guided build and lab, and the research it is based on.

PREF Preface: Why This Book, and How It Works

Why it was written and how to use it.

Part I · Chapters 1 to 3

From Data to Decisions

Main goal: Connect data to business performance, classify data correctly and organize information.

1 From Data to Business Performance

How does a business turn everyday activity into better results?

Let's Build: Trace one KPI back to the records that produce it.

Interactive lab: A KPI audit of a small business: three KPIs, the data each needs, and who decides with it.

2 Understanding Data

What kind of data is this, and what can I legitimately do with it?

Let's Build: Profile a dataset column by column: type, scale and allowed operations.

Interactive lab: Data Type Detective: classify and repair a messy business file, then draw one valid insight per column type.

3 Organizing Data and Information

How do we make data findable, reusable and trustworthy?

Let's Build: A data dictionary and a folder-and-tag scheme for the running case.

Interactive lab: Reorganize and document a shared-drive mess.

Part II · Chapters 4 to 6

Databases and SQL

Main goal: Read a relational database and answer business questions with SQL.

4 Databases and the Relational Model

Why do organizations keep data in related tables instead of files?

Let's Build: Explore a supplied case database in Supabase Studio and trace a relationship.

Interactive lab: Read an unfamiliar schema and predict the answers to five business questions before querying.

5 SQL Foundations

How do I ask one table a precise question?

Let's Build: Answer ten business questions on the running case.

Interactive lab: The same skills on the second case, with verification tasks.

6 SQL for Business Analysis

How do I combine tables into trustworthy business measures?

Let's Build: Build a small KPI report from joined tables.

Interactive lab: Find and fix a double-counted measure.

MID Integration Studio A (midterm)

An unnumbered review week between Parts II and III.

Cumulative review of Chapters 1 to 6, a practice exam, and **Project Checkpoint 1** : proposal and data plan.

> Based on: most of the courses reviewed run staged, multi-phase projects, so the project has graded checkpoints through the semester.

Part III · Chapters 7 to 11

Designing and Running Databases

Main goal: Design, normalize, build, test and secure a database of their own.

7 Database Design: From Requirements to ER Diagrams

How do business rules become a database design?

Let's Build: Draw the ERD for the running case from its business rules.

Interactive lab: Design an ERD for a new business process.

8 Normalization

How do we design out errors before they happen?

Let's Build: Normalize a flat spreadsheet to third normal form.

Interactive lab: Normalize a real business form and prove no data was lost.

9 Building and Testing a Database

How do we turn a design into a working, trustworthy database?

Let's Build: Build and load the running-case database in Supabase.

Interactive lab: Build the lab case from its ERD and run an integrity test suite.

10 Advanced SQL

How do analysts write queries that are reusable and easy to check?

Let's Build: A reusable analysis layer of views and CTEs.

Interactive lab: A monthly performance report with rankings and period-over-period change.

11 Database Administration, Security and Reliability

How do we keep business data safe, correct and available?

Let's Build: Add roles, row-level security and an audit trigger to the running case.

Interactive lab: A security and recovery plan for the lab case. Project Checkpoint 2: working database.

Part IV · Chapters 12 to 15

From Data to Insight

Main goal: Analyze results, visualize them, report in Power BI and work with warehouses and AI.

12 Business Analytics and Decision-Making

What does the data actually tell us, and how sure are we?

Let's Build: A SQL analysis that answers one management question.

Interactive lab: An analytical memo for the lab case.

13 Reporting and Data Visualization

How do we show results so people decide well?

Let's Build: Redesign a misleading dashboard.

Interactive lab: A one-page dashboard specification.

14 Power BI

How do we turn a database into an interactive business report?

Let's Build: A Power BI report on the running case.

Interactive lab: A Power BI report on the lab case, validated against SQL.

15 Big Data, Data Warehouses and AI

How do organizations work with data at scale, and where does AI fit?

Let's Build: A small star schema from the running case.

Interactive lab: Evaluate three AI answers about the lab data: accept, revise or reject.

Part V · Chapter 16 and Closing

Putting It to Work

Main goal: Deliver a capstone decision package and plan a path into data careers.

16 Capstone: From Data to Decisions

Can you take a real business problem all the way to a defensible decision?

Project Checkpoint 3: the final project, with a working database, analysis, dashboard and decision memo, and an individual demonstration.

END Closing: Where Data Can Take You

Where do these skills lead?

Appendices

A

Supabase and the In-Book SQL Editor

Setting up a Supabase project, using Supabase Studio, and running PostgreSQL inside the reader.

B

MySQL Route

Tested schemas, seed data and solutions for every SQL chapter, for courses built on MySQL.

C

Microsoft Access Route

Tables, relationships, visual queries, forms and reports on the same cases.

D

SQL Reference and Dialect Guide

PostgreSQL, MySQL, SQL Server and SQLite side by side.

E

Diagramming with Lucidchart

ER diagrams in Lucidchart, with diagrams.net and Mermaid as free alternatives.

F

Tableau Route

The visualization chapter's activities in Tableau.

G

Aa

Terms Treasury

The glossary of every key term in the book.

How students progress

• **Understand the data** Chapters 1 to 3: what data is, how it measures performance, and how to organize it.

• **Use a database** Chapters 4 to 6: query a supplied database before designing one.

• **Build one** Chapters 7 to 11: design, normalize, build and secure their own database.

• **Decide and defend** Chapters 12 to 16: turn results into dashboards and a decision memo they can explain and defend.

Two ways to use the book

Both routes come from the syllabus research: business-core courses stop at design and reporting, while MIS courses need the full database sequence.

Business-core data course	Chapters 1 to 8, then 12 to 14.
MIS database course	Chapters 1 and 4 to 16, with Chapters 2 and 3 as assigned reading.

How it is different

This book differentiates itself by teaching business students to use data to make decisions. Therefore, it differs from the main textbooks mainly by:

• **Starting with the business.** Chapter 1 opens with performance, goals and KPIs, before any database theory.

• **Teaching data before databases.** Data types, measurement scales and organizing information come first.

• **Using a database before designing one.** Students query a supplied database in Chapters 4 to 6, then design their own.

• **Ending with the decision.** Analytics, dashboards and a full Power BI chapter lead to a decision memo.

• **One SQL dialect, flexible tools.** SQL is taught in the PostgreSQL dialect, with a portable core and explicit dialect differences, with planned, tested MySQL and Access routes on the same cases.

• **Industry tools students can keep.** Supabase provides a hosted PostgreSQL project; Power BI supplies a reporting studio, with Tableau as an alternate route. Account limits, licensing and

classroom access need an agreed support plan.

• **Making AI a habit.** Students practice accepting, revising or rejecting AI-assisted work, with responsibility for checking the result.

• **Practice in every chapter.** A Let's Build, an interactive lab and an auto-graded quiz, all on the book's own cases.

• **Developed in class.** Version 1 was tested in summer 2026; version 2 is in use with 500 students this fall.

Advantages

The courseware already works in the classroom, and each chapter ties reading, guided practice, a lab and an auto-graded quiz to the same business cases.

Ready asset	Live, in its second version, and used by 500 students this semester. The textbook, labs, quizzes and instructor materials exist today, so the work ahead is revision and review, not a first draft.
Two course routes	One book serves the business-core data course and the MIS database course through two routes, from the same chapters.
Portfolio opportunity	No textbook dominated the reviewed collection, and the applied titles that serve it are several years old. This book can serve that course with current SQL, analytics and AI, and complement the deeper texts rather than compete with them.
Known partner	I already work with Cengage as a member of its Intro Computing Thought Leadership Board, and I have taught this course for seven years.
Additional opportunities	Adjacent courses such as accounting information systems and business analytics foundations, custom editions, and MySQL and Access routes that let current adopters keep the labs they already run.

Open questions

Five decisions to make together.

Positioning and content	Which course comes first, and how should the outline complement Cengage’s current titles?
Tools and access	Which PostgreSQL, MySQL and Access routes should be fully supported, and how should students access Power BI?

Platform fit	How should readings, assessments, labs and executable SQL fit MindTap, Brightspace and the live reader?
Running cases	Keep Grading Database and PetVax, or test a commercial case such as retail orders and revenue?
Review and development	What should reviewers receive, what criteria should they use, and what scope, support and milestones should guide development?

PDF research library

The earlier research, prospectuses, dashboards and presentation are part of the review package.

Document	What it contains	PDF	Original Drive file
Strategic insight dashboard	Earlier market / positioning exhibit	Open PDF ↗	Original ↗
Strategic analysis	Alternate file of the same two-page dashboard	Open PDF ↗	Original ↗
Full earlier market research	Market analysis presentation	Open PDF ↗	Original ↗
Publisher acquisition dashboard	Earlier commercial and courseware case	Open PDF ↗	Original ↗
Product prospectus	Earlier four-page prospectus	Open PDF ↗	Original ↗
Cengage pitch	Earlier strategic acquisition proposal	Open PDF ↗	Original ↗
Detailed product prospectus	Earlier three-page prospectus	Open PDF ↗	Original ↗
Author presentation	Thirteen-slide book overview	Open PDF ↗	Original ↗

Original PDFs are preserved unchanged and reflect earlier research dates, platform plans and positioning. The dashboard and strategic-analysis files are identical; they are retained as separate source files, not counted as independent studies. Use the findings above for the reconciled proposal.

Earlier interactive research pages

Exhibit	Open
2026 10 10 Prior Textbook Contents Audit	View exhibit ↗
Bitm330 Publisher Dashboard 1	View exhibit ↗
Bitm330 Publisher Dashboard	View exhibit ↗
Cengage Meeting Handout Updated	View exhibit ↗
Cengage Meeting Handout	View exhibit ↗
Mis Strategic Insight Dashboard	View exhibit ↗
Mis Textbook Advanced Phase	View exhibit ↗
Mis Textbook Blueprint	View exhibit ↗
Mis Textbook Market Analysis Infographic 1	View exhibit ↗
Mis Textbook Market Analysis Infographic	View exhibit ↗

Suggested next steps

• **Cement the relationship.** Agree on the course opportunity and whether to move forward.

• **Review the outline together.** Settle the audience, scope, tools, cases and sample-unit requirements.

• **Develop and submit.** Set review milestones, responsibilities and the manuscript submission plan.

Let’s cement this deal.

Using Data to Drive Business Performance

Explore the existing courseware and the materials behind the proposed next edition.

Open the live reader ↗

Read the two-page proposal ↗

References

Market, employment and curriculum references

References

• National Center for Education Statistics. (2023). Digest of Education Statistics, Table 318.20: Degrees conferred by field of study, 2021-22. Link

• Fustos, J., et al. (2017). Do students get enough information systems education in the business core? Study of 61 business schools. Link

• Data USA. (2026). Management information systems and services (IPEDS completions, 2024). Link

• Hwang, D., Ma, Z., & Wang, M. (2015). The information systems core: A study from the perspective of IS core curricula in the U.S. Information Systems Education Journal, 13(6), 27-34. Link

• Zheng, Y., Hameed, T., Lavoie, R., & Sendall, P. (2021). An overview of current business analytics programs across US AACSB schools. Issues in Information Systems, 22(2), 306-317. Link

• Yang, S. C. (2023). The master's program in information systems: A survey of core curricula in AACSB-accredited business schools in the United States. Journal of Information Systems Education, 34(4), 430-440. Link

• Data USA. (2026). Accounting (IPEDS completions, 2024). Link

• AACSB International. (2026, September). Accreditation announcement, September 2026. Link

• 365 Data Science. (2026). The data analyst job market in 2026: Research on 1,000 job postings. Link

• U.S. Bureau of Labor Statistics. (2026). Occupational Outlook Handbook: Data scientists. Link

• U.S. Bureau of Labor Statistics. (2026). Occupational Outlook Handbook: Database administrators and architects. Link

• World Economic Forum. (2025). The Future of Jobs Report 2025. Link

• Cengage. Concepts of Database Management, 10th edition (2021). Link

Manuscript, textbook, platform and original-cover references

[S1]

Collected syllabus archive and manifest ↗

63 documents; 31 institutions; archived October 1, 2026. Mixed course levels and dates.

[S2]

Database-platform evidence model and report ↗

October 17, 2026 audit. Role-coded platform evidence, 54 selected course families; counts recalculated for this package.

[S3]

Curriculum audit ↗

October 1, 2026 document-level topic summaries and original-source references; used as a discovery aid and checked against archived course text.

[S4]

Textbook adoption and contents audit ↗

October 10, 2026 report: 60 named books/course texts. Historical school-level mentions, not market shares. Broader contents catalogue retained as prior research.

[S5]

Stabilized 14-chapter proposal ↗

Earlier synthesis, stored in the outline-comparison folder. This package proposes a refinement; it does not replace the manuscript outline.

[S6]

Current book repository ↗

Read-only inspection of main, commit 92d7a6ea04b96b67ef983aa22d054ab0bb9ad046; current chapter activation metadata checked.

[S7]

Inactive book-2027 candidate ↗

16-chapter migration candidate with separate core/extension routes. Its README explicitly states that it is inactive and is not published by the reader.

[S8]

Dima Publishing design and book description ↗

Publisher design reference and earlier public description. Cover for this revision comes from the current book website instead; see C1.

[S9]

Brightlab evaluation design document ↗

Prototype description, not validation of fairness, reliability, costs or deployability. Contradictory partial-credit claims require resolution.

[P1]

Cengage: Database Systems, 14th edition ↗

Coronel and Morris. Uploaded copy supplies the historical chapter sequence; official product page documents interactive MySQL labs and other MindTap resources.

[P2]

Cengage: Database Systems, 15th edition ↗

Coronel and Morris. Official current listing checked October 25, 2026; detailed sequence is retained from the earlier contents audit. Recent edition moves SQL ahead of the formal design sequence.

[P3]

Cengage: Concepts of Database Management, 10th edition ↗

Friedrichsen, Ruffolo, Monk, Starks, Pratt and Last. Official publisher search results provide the nine-module outline and Access practice description.

[P4]

Pearson: Database Concepts, 10th edition ↗

Kroenke, Auer, Vandenberg and Yoder. Current publisher educator page; the uploaded PDF is the older ninth edition, not the tenth.

[P5]

Pearson: Modern Database Management, 14th edition ↗

Hoffer, Topi, Venkataraman and Bala. Official contents and description checked October 25, 2026; explicitly covers analytics, cloud and AI.

[L1]

U.S. Bureau of Labor Statistics: Data Scientists ↗

Current Occupational Outlook Handbook read October 25, 2026. Projects 35% employment growth, 2025-2035; all occupations 3%.

[L2]

U.S. Bureau of Labor Statistics: Database Administrators and Architects ↗

Current Occupational Outlook Handbook read October 25, 2026. Architects 9%; administrators 0%; combined 4%, 2025-2035. Rounded percentages.

[L3]

World Economic Forum: Future of Jobs Report 2025 ↗

Employer expectations for 2025-2030: analytical thinking essential to about seven in ten responding companies; AI and big data lead skills expected to grow. Global employer survey, not U.S. hiring requirements.

[T1]

PGlite: What is PGlite? ↗

Official documentation: embedded PostgreSQL compiled to WebAssembly; runs in the browser, with optional local persistence. It is not itself a hosted cloud database.

[T2]

Supabase: Database overview ↗

Official documentation: each project provides a PostgreSQL database. Hosted operations, accounts, permissions and product limits require a separate teaching plan.

[T3]

Microsoft: Get Power BI Desktop ↗

Current installation documentation, checked October 25, 2026. Windows desktop authoring creates an access requirement for Mac users.

[M1]

NCES: Undergraduate Degree Fields ↗

2021-22: 375,400 business bachelor’s degrees, 19% of bachelor’s awards. Broad audience context; degree completions are not database-course enrollments.

[M2]

Cengage: Database Systems, 15th edition, U.S. product listing ↗

Pricing retrieved October 25, 2026: eBook US$71.99 through January 22, 2027 or US$98.99 through July 21, 2027; paperback US$272.95; MindTap advertised from US$149.00. Prices and access dates can change.

[C1]

Original cover on the current book website ↗

Existing animated cover URL verified in the current repository’s HomePage component. Used unchanged, with a still-image fallback for reduced motion.